THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. In science, a model is best described as: 1

(a) an untested guess
(b) a final truth that never changes
(c) an exact copy of the real object
(d) a simplified representation of a system that keeps what matters for the question

2. A pure substance consists of: 1

(a) two or more substances in any ratio
(b) a single kind of particle with fixed composition
(c) only gases
(d) only metals

3. The law of conservation of mass was proposed by: 1

(a) John Dalton **(b)** Antoine Lavoisier
(c) Joseph Proust **(d)** J. J. Thomson

4. The sub-atomic particle discovered by J. J. Thomson was the: 1

(a) nucleus **(b)** electron **(c)** neutron **(d)** proton

5. A runner completes one full lap of a 400 m circular track. Her displacement at the end is: 1

(a) 200 m **(b)** 800 m **(c)** 0 **(d)** 400 m

6. A body continues in its state of rest or of uniform motion in a straight line unless acted on by an external force. This is: 1

(a) Newton's third law **(b)** Newton's second law
(c) Newton's first law **(d)** the law of gravitation

7. In science, work is done when: 1

(a) a person thinks hard
(b) a force acts on a body and the body is displaced in the direction of the force
(c) a person holds a heavy load at rest
(d) a force acts but there is no displacement

8. Sound is produced by: 1

(a) stationary bodies **(b)** vibrating bodies
(c) heat only **(d)** light

9. The cell was first observed and named by: 1

(a) Anton van Leeuwenhoek **(b)** Rudolf Virchow
(c) Robert Hooke **(d)** Robert Brown

10. A group of similar cells performing a specific function is called a: 1

(a) organ **(b)** organelle **(c)** tissue **(d)** organism

11. The variety of living organisms found on the Earth is called: 1

(a) heredity **(b)** taxonomy **(c)** biodiversity **(d)** ecology

12. Reproduction in which a single parent produces offspring identical to itself is: 1

(a) fertilisation **(b)** asexual reproduction
(c) pollination **(d)** sexual reproduction

13. The four spheres of the Earth system are the geosphere, hydrosphere, atmosphere and: 1

(a) lithosphere only **(b)** troposphere **(c)** biosphere **(d)** stratosphere

14. A rough estimate is useful when: 1

(a) an exact value is required
(b) no data are available at all
(c) we want to avoid reasoning
(d) an approximate answer is enough for a decision

15. A suspension is: 1

(a) a pure substance
(b) a clear solution
(c) a compound
(d) a heterogeneous mixture whose particles settle on standing

16. The atomicity of phosphorus (P_4) is: 1

(a) 1 **(b)** 8 **(c)** 2 **(d)** 4

17. The atomic number of an element is equal to the number of: 1

(a) neutrons **(b)** protons
(c) valence electrons only **(d)** nucleons

18. The slope of a velocity-time graph gives: 1

(a) force **(b)** distance **(c)** acceleration **(d)** speed

19. According to Newton's third law, action and reaction forces: 1

(a) are unequal
(b) act on the same body
(c) cancel each other on the same body
(d) are equal in magnitude, opposite in direction and act on different bodies

20. The work-energy theorem states that the work done on a body is equal to the change in its: 1

(a) momentum **(b)** mass **(c)** kinetic energy **(d)** potential energy

21. The loudness of a sound depends mainly on its: 1

(a) wavelength **(b)** speed **(c)** amplitude **(d)** frequency

22. A cell without a well-defined nucleus is a: 1

(a) animal cell **(b)** eukaryotic cell **(c)** plant cell **(d)** prokaryotic cell

23. The tissue that makes the husk of a coconut is the: 1

(a) sclerenchyma **(b)** phloem **(c)** parenchyma **(d)** collenchyma

24. The simplest plants with a thallus-like body are: 1

(a) ferns **(b)** mosses **(c)** gymnosperms **(d)** algae

25. The male reproductive part of a flower is the: 1

(a) carpel **(b)** sepal **(c)** petal **(d)** stamen

26. The surface that reflects the most solar radiation is: 1

(a) ocean **(b)** forest **(c)** fresh snow **(d)** dark soil

27. A 'law' of science and a 'theory' differ because: 1

(a) a law describes what happens while a theory explains why
(b) a law is a guess
(c) a theory is untested
(d) they are the same

28. In paper chromatography the pencil line is drawn at the bottom because: 1

(a) pencil does not dissolve in the solvent **(b)** it looks neat
(c) pencil is faster **(d)** ink is better

29. An anion is a: 1

(a) positively charged ion **(b)** neutral particle
(c) molecule **(d)** negatively charged ion

30. The valency of an element with 7 valence electrons is: 1

(a) 1 **(b)** 7 **(c)** 6 **(d)** 8

31. The equation v squared = u squared + 2as relates: 1

(a) velocity and time only **(b)** velocity, acceleration and displacement
(c) force and mass **(d)** work and time

32. The weight of a body on the Moon compared to that on Earth is about: 1

(a) the same **(b)** one-sixth **(c)** six times **(d)** zero

33. A simple machine is a device that: 1

(a) does more work than the input
(b) needs no effort
(c) makes work easier by changing the magnitude or direction of force
(d) creates energy

34. The minimum distance of the reflecting surface to hear an echo in air (v = 344 m/s) is about: 1

(a) 17.2 m **(b)** 344 m **(c)** 34.4 m **(d)** 1.72 m

35. The endoplasmic reticulum with ribosomes attached is the: 1

(a) smooth ER **(b)** vacuole **(c)** Golgi **(d)** rough ER

36. Animal tissue that covers or lines the body surfaces is: 1

(a) epithelial tissue **(b)** muscular tissue
(c) connective tissue **(d)** nervous tissue

37. Sponges belong to the phylum: 1

(a) Cnidaria **(b)** Mollusca **(c)** Porifera **(d)** Annelida

38. After fertilisation, the ovary develops into a: 1

(a) flower **(b)** fruit **(c)** anther **(d)** seed

39. The ozone layer is present in the: 1

(a) geosphere **(b)** troposphere **(c)** mesosphere **(d)** stratosphere

40. Chemists draw atoms as spheres and bonds as sticks. This is an example of: 1

(a) a law **(b)** a unit **(c)** an estimate **(d)** a model

41. Which of the following is a pure substance? 1

(a) distilled water **(b)** milk **(c)** air **(d)** sea water

42. The law of constant proportions was proposed by: 1

(a) Lavoisier **(b)** Rutherford **(c)** Dalton **(d)** Joseph Proust

43. The charge on an electron is: 1

(a) variable **(b)** negative **(c)** zero **(d)** positive

44. The distance travelled by a body is always: 1

(a) less than displacement
(b) equal to velocity
(c) negative
(d) greater than or equal to the magnitude of displacement

45. Inertia of a body depends on its: 1

(a) colour **(b)** speed only **(c)** volume **(d)** mass

46. The SI unit of work is the: 1

(a) newton **(b)** pascal **(c)** watt **(d)** joule

47. Sound cannot travel through: 1

(a) a vacuum **(b)** water **(c)** steel **(d)** air

48. Cell theory was proposed by: 1

(a) Darwin and Wallace **(b)** Watson and Crick
(c) Schleiden and Schwann **(d)** Hooke and Brown

49. Muscles attached to bones and used in running and writing are: 1

(a) cardiac muscles **(b)** striated (voluntary) muscles
(c) smooth muscles **(d)** involuntary branched muscles

50. Fish breathe through: 1

(a) lungs **(b)** trachea **(c)** skin only **(d)** gills

END OF QUESTION PAPER **Total: 50 Marks**

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